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Master Degree in Music and Acoustic Engineering
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Report of the Homeworks

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1

Homework 1

1.1 Weak Form and FE Discretization

In this chapter, the weak formulation and the corresponding finite element (FE) discretization of the boundary value problem introduced in Homework 1 are derived.

1.1.1 Strong Formulation

The one-dimensional wave propagation problem is considered:

$$\begin{cases} \rho(x)\omega^2 u(x) + \dfrac{\partial}{\partial x}\left(\mu(x)\dfrac{\partial u(x)}{\partial x}\right) = f \quad x \in (0, L) \\ \mu(0)\left.\dfrac{\partial u(x)}{\partial x}\right|_{x=0} = g_N \\ u(L) - \sqrt{\dfrac{\mu(L)}{\rho(L)}}\dfrac{i}{\omega}\left.\dfrac{\partial u(x)}{\partial x}\right|_{x=L} = g_A \end{cases} \tag{1.1}$$

In this report, the following notational simplifications are adopted:

- $u = u(x)$ and, analogously, for all functions depending on time;
- $u' = u'(x) = \frac{\partial u(x)}{\partial x}$ and, analogously, for all spatial derivatives;
- $u'(0) = \left.\frac{\partial u(x)}{\partial x}\right|_{x=0}$ and, analogously, for derivatives calculated in $x = L$.

The second boundary condition is an impedance (or absorbing) condition. It is useful to rewrite it in order to obtain $u'(L)$ explicitly. From the relation above, it follows that:

$$u'(L) = -i\omega\sqrt{\frac{\rho(L)}{\mu(L)}}\left(u(L) - g_A\right)$$

For convenience, the impedance coefficient is introduced as

$$\beta = \sqrt{\rho(L)\mu(L)}$$

1.1.2 Weak Formulation

To obtain the weak formulation, the strong form of the equation is multiplied by a test function $v(x) \in H^1(0, L)$ and integrated over $(0, L)$:

$$\int_0^L \rho\omega^2 uv dx + \int_0^L \frac{d}{dx}\left(\mu u'\right) v dx = \int_0^L f v dx$$

The second term is then integrated by parts:

$$\int_0^L \frac{d}{dx}(\mu u')v dx = [\mu u' v]_0^L - \int_0^L \mu u' v' dx$$

Substituting into the previous expression yields:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx + [\mu u' v]_0^L = \int_0^L f v dx$$

The boundary contributions are incorporated using the boundary conditions. At $x = 0$, it holds that:

$$\mu(0)u'(0)v(0) = g_N v(0) \quad \Longrightarrow \quad -\mu(0)u'(0)v(0) = -g_N v(0)$$

At $x = L$, using $u'(L)$ from the impedance condition, one obtains:

$$\mu(L)u'(L)v(L) = -i\omega\beta u(L)v(L) + i\omega\beta g_A v(L)$$

Collecting all terms, the weak formulation becomes:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx - i\omega\beta u(L)v(L) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L) \tag{1.2}$$

It is convenient to identify the mass term, stiffness term and the contributions arising from the boundary conditions. The bilinear forms are defined as:

$$\mathcal{M}(u,v) = \int_0^L \rho u v dx, \quad \mathcal{A}(u,v) = \int_0^L \mu u' v' dx + i\omega\beta u(L)v(L)$$

and the linear functional as:

$$\mathcal{F}(v) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L)$$

Thus, the weak formulation can be written compactly as:

$$\omega^2 \mathcal{M}(u,v) - \mathcal{A}(u,v) = \mathcal{F}(v) \quad \forall v \in H^1(0,L) \tag{1.3}$$

1.1.3 Finite Element Discretization and Algebraic System

A partition of the interval $(0, L)$ into N elements is introduced and the finite dimensional space $V_h \subset H^1(0,L)$, spanned by continuous, piecewise linear basis functions $\{\varphi_j\}_{j=1}^N$, is considered. The solution and the test function are approximated as:

$$u_h = u_h(x) = \sum_{j=1}^N u_j(x)\varphi_j(x), \quad v_h = v_h(x) = \varphi_i(x)$$

Substituting u_h and v_h into the equation (1.2) yields the discrete system. The entries of the mass matrix, stiffness matrix and load vector are then defined.

- Mass matrix:

$$M_{ij} = \int_0^L \varphi_i \varphi_j dx$$

- Stiffness matrix and impedance contribution:

$$A_{ij} = \int_0^L \mu \varphi_i' \varphi_j' dx + i\omega\beta\varphi_i(L)\varphi_j(L)$$

  Note that $\varphi_i(L) \neq 0$ is only for the basis function associated with the last node of the mesh. Therefore, the impedance term modifies only the entry A_{NN}.
- Load vector:

$$F_i = \int_0^L f\varphi_i dx + g_N\varphi_i(0) - i\omega\beta g_A\varphi_i(L)$$

  Here $\varphi_i(0)$ is nonzero only for the basis function associated with the first node, while $\varphi_i(L)$ is nonzero only for the last one.

Collecting all contributions, the linear system is obtained:

$$\omega^2 \mathbf{M}\underline{u} - \mathbf{A}\underline{u} = \underline{F} \tag{1.4}$$

which corresponds to the discretized weak formulation and matches the form requested in the homework assignment. This completes the derivation of the weak formulation and its finite element discretization using linear basis functions.

1.2 MATLAB Solver

This section describes the MATLAB implementation used to solve the algebraic system derived in section 1.1 and introduces a verification procedure to assess the correctness of the finite element (FE) code. In particular, the Method of Manufactured Solutions (MMS) is adopted, which consists of prescribing an exact solution and deriving the corresponding forcing terms and boundary data. This provides a controlled test case where the numerical approximation can be directly compared against a known reference.

1.2.1 Verification Test Setup

Considering the one-dimensional domain and constant physical parameters

$$\Omega = (0, 1), \qquad L = 1, \qquad \rho = 1, \qquad \mu = 1$$

For the verification test, the manufactured exact solution is imposed

$$u_{\text{ex}}(x) = \sin(\omega x), \qquad \omega = 5\pi$$

Internal forcing term Starting from the strong form in equation (1.1), the governing equation can be written as

$$\rho\omega^2 u(x) + \frac{d}{dx}\left(\mu u'(x)\right) = f(x) \tag{1.5}$$

where $u' = \frac{du}{dx}$. Computing the derivatives of the manufactured solution yields.

$$u_{\text{ex}}'(x) = \omega\cos(\omega x), \qquad u_{\text{ex}}''(x) = -\omega^2\sin(\omega x)$$

Since μ is constant, $\frac{d}{dx}(\mu u') = \mu u''$. Substituting u_{ex} into equation (1.5) with $\rho = \mu = 1$ yields

$$f(x) = \omega^2\sin(\omega x) + u_{\text{ex}}''(x) = \omega^2\sin(\omega x) - \omega^2\sin(\omega x) = 0$$

Therefore, the internal source term vanishes:

$$f(x) = 0 \tag{1.6}$$

Neumann condition The Neumann boundary condition in equation (1.1) at $x = 0$ reads

$$\mu(0)u'(0) = g_N \tag{1.7}$$

Using $u'_{\text{ex}}(x) = \omega\cos(\omega x)$ yields

$$u'_{\text{ex}}(0) = \omega\cos(0) = \omega, \qquad \mu(0) = 1$$

hence

$$g_N = \omega = 5\pi \tag{1.8}$$

Impedance condition The impedance boundary condition in equation (1.1) at $x = L$ is

$$u(L) - \sqrt{\frac{\mu(L)}{\rho(L)}}\frac{i}{\omega}u'(L) = g_A \tag{1.9}$$

Since $\rho(L) = \mu(L) = 1$, the impedance coefficient simplifies to

$$\sqrt{\frac{\mu(L)}{\rho(L)}} = 1$$

Evaluating the manufactured solution at $L = 1$ gives

$$u_{\text{ex}}(1) = \sin(5\pi) = 0, \qquad u'_{\text{ex}}(1) = 5\pi\cos(5\pi) = -5\pi$$

Substituting into the boundary condition yields

$$g_A = u_{\text{ex}}(1) - \frac{i}{\omega}u'_{\text{ex}}(1) = 0 - \frac{i}{5\pi}(-5\pi) = i$$

Therefore, the impedance boundary datum is

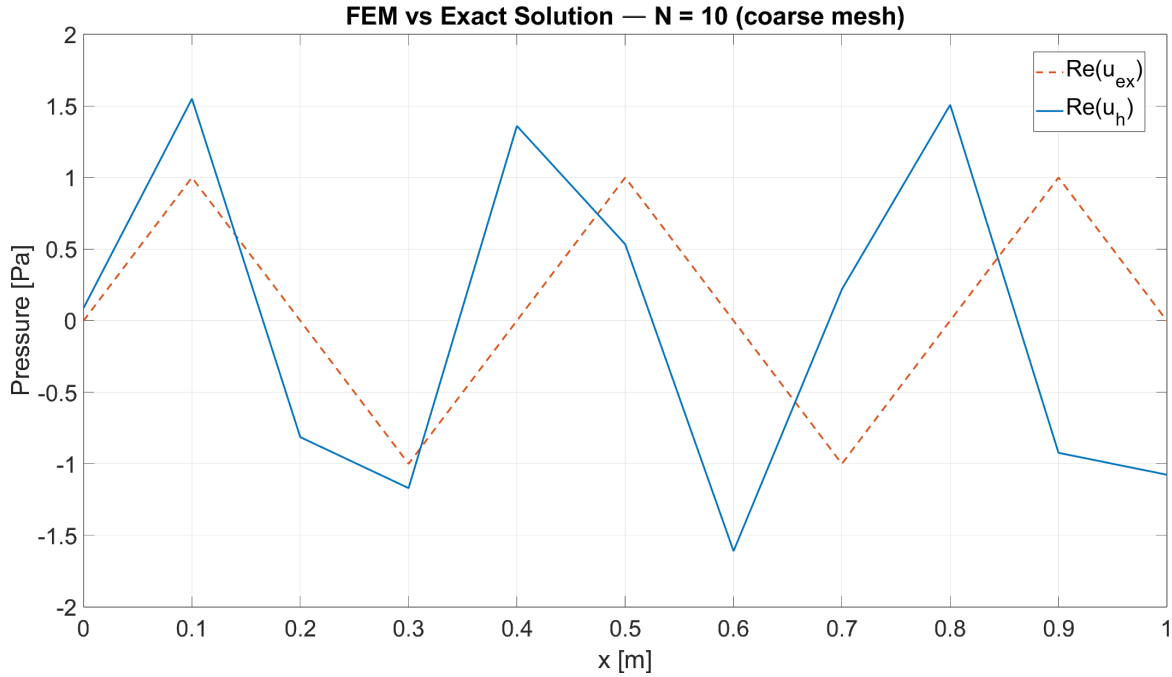
$$g_A = i \tag{1.10}$$

1.2.2 Numerical Results

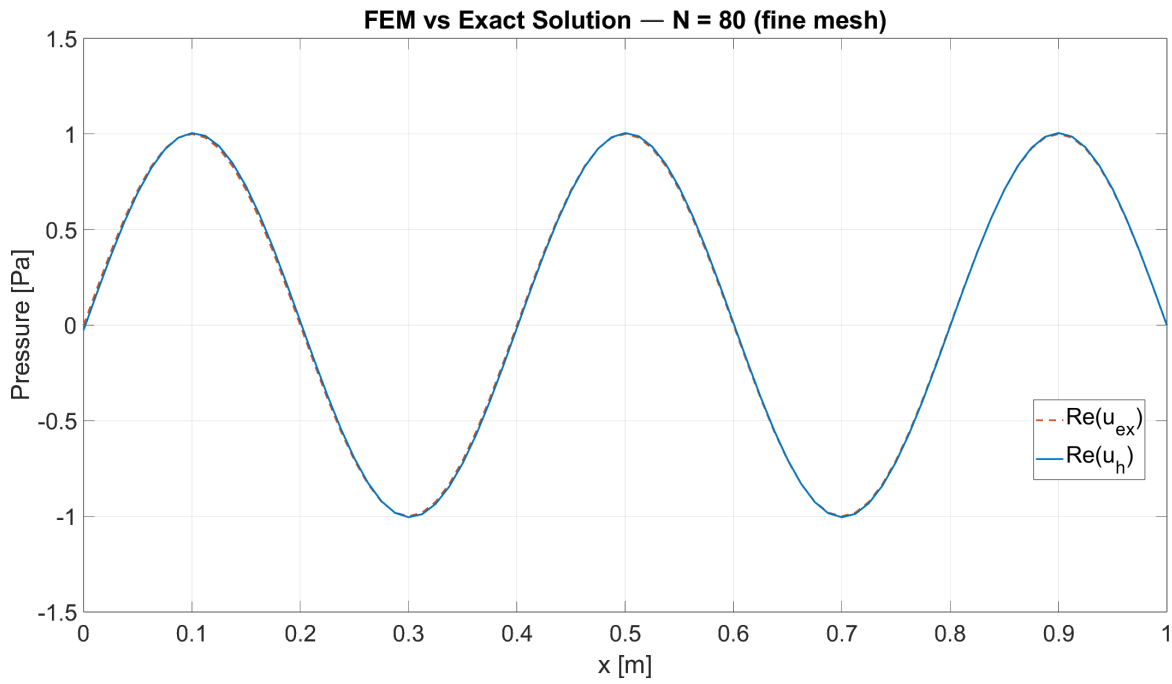
The algebraic system (1.4) is assembled and solved in MATLAB for a sequence of uniform meshes with $N \in \{10,\ 20,\ 40,\ 80\}$ elements. Since $\omega = 5\pi$, the wavelength of the exact solution is

$$\lambda = \frac{2\pi}{\omega} = \frac{2}{5} = 0.4\,\text{m}$$

so the domain $(0, 1)$ contains exactly 2.5 full wavelengths. On the coarsest mesh ($N = 10$, $h = 0.1$), each wavelength is covered by only $\lambda/h = 4$ elements, which is insufficient to resolve the oscillatory behavior of u_{ex}. On the finest mesh ($N = 80$, $h = 0.0125$), the resolution rises to 32 elements per wavelength, yielding an accurate approximation. Figure 1.1 compares the real part of the FE solution u_h with the exact solution u_{ex} on the coarse and fine meshes respectively. As expected, the coarse-mesh solution captures the correct frequency but with a noticeable phase and amplitude error, while the fine-mesh solution overlaps the exact curve almost perfectly.



(a) On the coarse mesh ($N = 10$, $h = 0.1$). The four elements per wavelength are insufficient to resolve the sinusoidal shape, resulting in a visible phase shift and amplitude overshoot



(b) On the fine mesh ($N = 80$, $h = 0.0125$). With 32 elements per wavelength the numerical and exact curves are virtually indistinguishable

Figure 1.1: Real part of the FE pressure field compared with the exact solution $u_{\text{ex}} = \sin(\omega x)$

1.2.3 Convergence Analysis

To quantify the approximation quality, the $L^2(0,1)$ error is computed for each mesh via numerical quadrature:

$$\|u - u_h\|_{L^2(0,L)} = \left(\int_0^L |u_{\text{ex}}(x) - u_h(x)|^2 dx \right)^{1/2}$$

The theoretical a-priori estimate for continuous piecewise-linear ($\mathbb{P}_1$) Galerkin approximations of elliptic problems guarantees

$$\|u - u_h\|_{L^2(0,L)} \leq Ch^2 |u|_{H^2(0,L)} \tag{1.11}$$

so one expects a convergence rate of order 2 in the mesh size h. Table 1.1 reports the computed L^2 errors and the corresponding numerical convergence rates p, estimated from consecutive mesh pairs as

$$p = \frac{\log(e_i/e_{i+1})}{\log(h_i/h_{i+1})}$$

N	h	$\|u - u_h\|_{L^2}$	Rate p
10	0.1000	8.21×10^{-1}	—
20	0.0500	2.05×10^{-1}	2.04
40	0.0250	5.12×10^{-2}	2.21
80	0.0125	1.28×10^{-2}	3.05

Table 1.1: L^2 errors and estimated convergence rates for the MMS verification test with $u_{\text{ex}} = \sin(5\pi x)$ and NI boundary conditions

The numerical rates confirm the expected $\mathcal{O}(h^2)$ convergence, consistent with the theoretical estimate (1.11) and with the use of $\mathbb{P}_1$ basis functions. The log-log convergence plot in figure 1.2 shows that the error decreases along a straight line of slope 2, parallel to the reference $\mathcal{O}(h^2)$ dashed line. This agreement between theory and numerics validates the correctness of the MATLAB implementation.

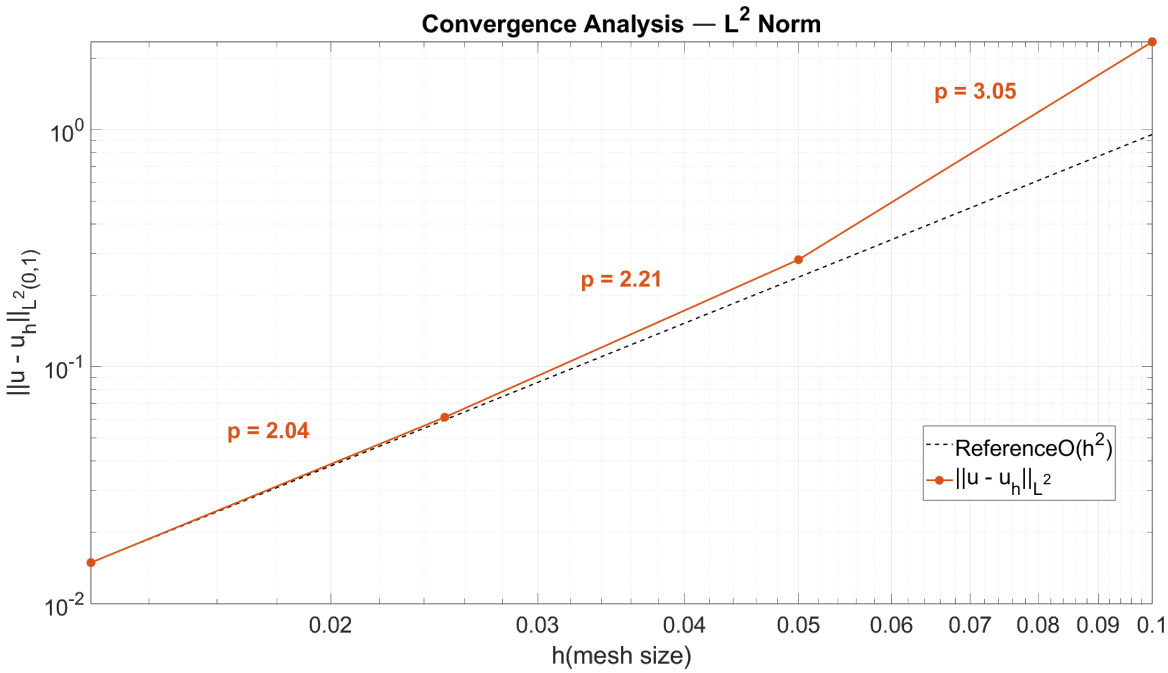


Figure 1.2: Log-log convergence plot of the L^2 error $\|u - u_h\|_{L^2(0,1)}$ as a function of the mesh size h for the MMS test with $u_{\text{ex}} = \sin(5\pi x)$. The dashed line represents the reference slope $\mathcal{O}(h^2)$ and the annotated rates confirm quadratic convergence for $\mathbb{P}_1$ elements

1.3 Wave Reflection at an Impedance Jump

The propagation of a time-harmonic field in a heterogeneous one-dimensional medium is considered, with a discontinuity in the material properties located at mid-domain. The main effect of such discontinuity is the generation of a partially reflected wave due to an impedance mismatch.

1.3.1 Problem Setup

The spatial domain is defined as $\Omega = (0, 1)$ with constant density $\rho(x) = 1$. The stiffness is assumed piecewise constant and discontinuous at $x = 0.5$, according to

$$\mu(x) = \begin{cases} 4, & x \leq 0.5 \\ 1, & x > 0.5 \end{cases} \tag{1.12}$$

No volumetric source term is applied, hence $f(x) = 0$. The excitation is imposed at the left boundary through a Neumann condition $\mu(0)u'(0) = g_N$ with $g_N = 1$, whereas at the right boundary a perfectly absorbing condition is prescribed by setting $g_A = 0$. This choice prevents spurious reflections at $x = 1$, so that the observed reflected component is generated only at the internal interface $x = 0.5$.

1.3.2 Mechanical Impedance of the Two Media

For one-dimensional wave propagation, reflection and transmission at an interface are governed by the characteristic (mechanical) impedance of the medium, defined as

$$Z = \sqrt{\rho\mu} \tag{1.13}$$

Since $\rho = 1$ throughout Ω, the impedance depends only on μ. In the left subdomain $x \leq 0.5$ it follows that $Z_1 = \sqrt{1 \cdot 4} = 2$, while in the right subdomain $x > 0.5$ one obtains $Z_2 = \sqrt{1 \cdot 1} = 1$. The corresponding phase velocities are $c_1 = \sqrt{\mu_1/\rho} = 2$ and $c_2 = \sqrt{\mu_2/\rho} = 1$ respectively.

1.3.3 Reflection Coefficient at the Interface

When a right-going incident wave traveling in medium 1 reaches the interface with medium 2, continuity of displacement u and traction $\mu u'$ implies the formation of both a reflected and a transmitted component. The reflection coefficient for incidence from medium 1 to medium 2 is given by

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2} \tag{1.14}$$

Substituting $Z_1 = 2$ and $Z_2 = 1$ yields

$$R = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

Therefore, a partial reflection is expected at $x = 0.5$ with positive sign, i.e., without phase inversion.

1.3.4 Expected Qualitative Behavior in Amplitude and Phase

Distinct qualitative behaviors are expected on the two sides of the interface. In the left region $x < 0.5$, the superposition of the incident and reflected waves produces an interference pattern, corresponding to a partially standing-wave regime. As a consequence, oscillations are expected in both the amplitude profile $|u(x)|$ and the phase $\arg(u(x))$, which will be irregular and non-monotone. In the right region $x > 0.5$, the transmitted wave propagates toward $x = 1$. Due to the perfectly absorbing condition $g_A = 0$, no reflection is generated at the right boundary and the solution resembles a single traveling wave. Hence, the amplitude is expected to be comparatively smooth and the phase to grow linearly and monotonically with slope $\omega/c_2 = \omega$. At the interface $x = 0.5$, a discontinuity in the spatial derivative of the phase is expected, reflecting the change in phase velocity from $c_1 = 2$ to $c_2 = 1$: since the phase slope of a traveling wave is ω/c, the slope in the right subdomain is twice that in the left subdomain for a purely traveling wave.

1.3.5 Numerical Results

The problem is solved numerically using the MATLAB solver described in chapter 1.1, with $N = 200$ elements to ensure sufficient resolution across both subdomains. The stiffness coefficient is implemented as a piecewise function evaluated at the quadrature nodes of each element. Figure 1.3 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ of the computed solution.

1.3.6 Physical Interpretation

The numerical results are in full agreement with the analytical predictions derived in the previous sections. Three distinct regions can be identified in the solution.

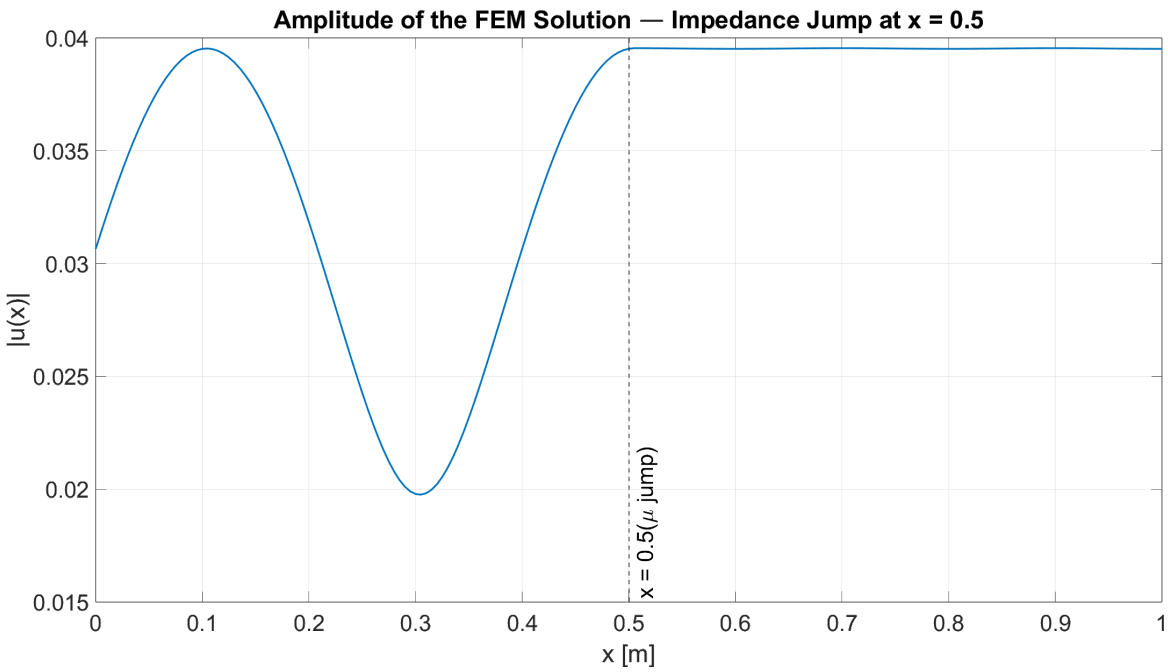
Left subdomain $x < 0.5$ — partial standing wave The amplitude profile $|u(x)|$ exhibits oscillations with alternating local maxima and minima, which are the signature of the interference between the rightward-traveling incident wave and the leftward-traveling reflected wave generated at the interface. The phase is irregular and non-monotone, consistently with the superposition of two counter-propagating components. The reflection coefficient $R = 1/3$ implies that approximately 11% of the incident energy is reflected back into the left subdomain, producing a moderately pronounced standing-wave pattern.

Right subdomain $x > 0.5$ — traveling wave Beyond the interface, the amplitude is smooth and the phase grows linearly with slope $\omega/c_2 = \omega$, confirming the presence of a single right-going traveling wave. The absorbing boundary condition at $x = 1$ effectively prevents any reflection from the right end, so no additional interference pattern appears in this region.

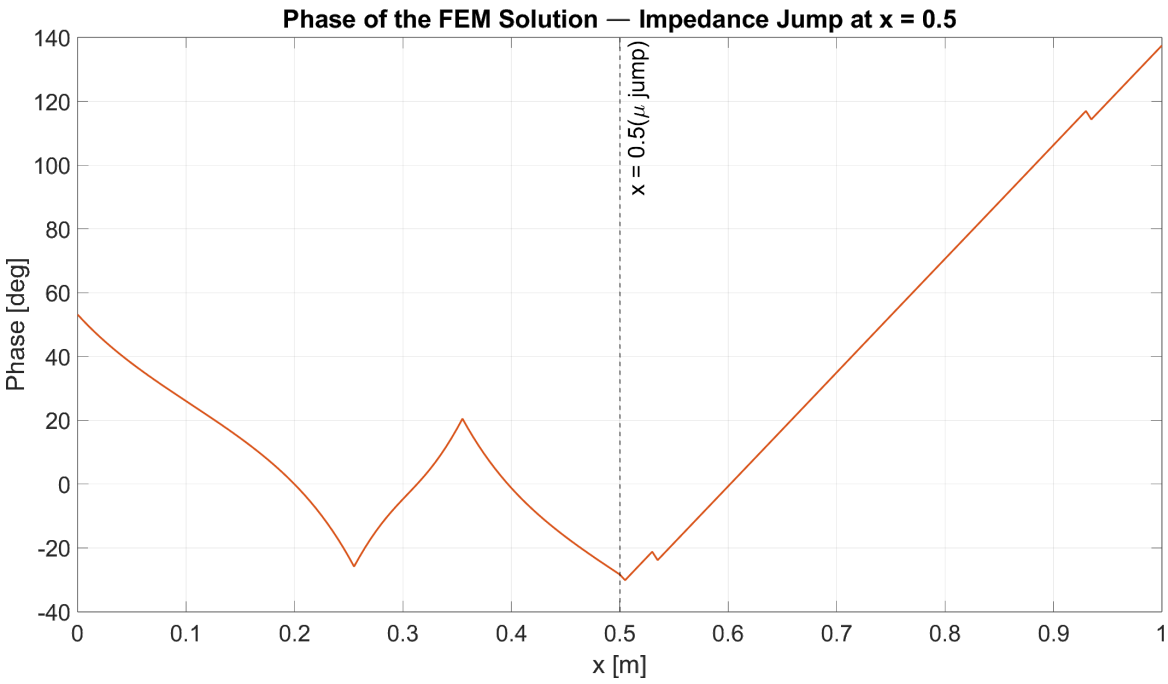
Interface $x = 0.5$ — phase velocity change A clear kink in the phase is visible at the interface. Since the phase slope of a traveling wave equals ω/c, the ratio of the slopes on the two sides of the interface is

$$\frac{(\omega/c_2)}{(\omega/c_1)} = \frac{c_1}{c_2} = \frac{2}{1} = 2$$

confirming that the phase advances twice as fast per unit length in the right medium, consistently with $c_2 = 1 < c_1 = 2$.



(a) Amplitude $|u(x)|$ of the numerical solution. The oscillatory pattern in the left subdomain ($x < 0.5$) is the signature of the standing-wave component generated by the partial reflection at the interface, while the smoother decay in the right subdomain ($x > 0.5$) corresponds to the purely transmitted wave



(b) Unwrapped phase $\arg(u(x))$ of the numerical solution. The irregular, non-monotone behavior for $x < 0.5$ is due to the interference between the incident and reflected waves, while the linear monotone growth for $x > 0.5$ is consistent with a single right-going traveling wave. The kink at $x = 0.5$ reflects the change in phase velocity between the two media

Figure 1.3: Amplitude and unwrapped phase of the FE solution for the impedance-jump problem with μ as in equation (1.12), $\rho = 1$, $g_N = 1$, $g_A = 0$ and $\omega = 5\pi$

1.4 Numerical Dispersion Analysis

It is well known that spatial discretization introduces a dispersion error: the discrete solution does not propagate with the same phase velocity as the continuous (physical) wave. In

the frequency domain, this phenomenon is conveniently quantified by comparing the continuous wavenumber k with the discrete (numerical) wavenumber k_h, associated with the finite-dimensional approximation. The resulting discrepancy represents the numerical dispersion of the method.

1.4.1 Eigenvalue Problem Setting

To isolate dispersion effects from external excitations, the homogeneous problem is considered by setting $f = 0$ and imposing homogeneous boundary data. In particular, a homogeneous Neumann condition is prescribed at the left boundary, namely $g_N = 0$, which implies $\mu u'(0) = 0$, while a homogeneous Dirichlet condition is imposed at the right boundary, $u(L) = 0$. Under these assumptions, the Helmholtz model reduces to an eigenvalue problem for the spatial operator.

1.4.2 Continuous and Discrete Formulations

In the continuous setting and for constant coefficients $\rho = \mu = 1$, the eigenmodes satisfy the classical second-order equation

$$-u''(x) = k^2 u(x) \tag{1.15}$$

where k^2 denotes the exact eigenvalues of the continuous problem and k is the corresponding continuous wavenumber. By applying the finite element discretization derived in section 1.1 and restricting the solution to the discrete space, the homogeneous algebraic system assumes the form of a generalized eigenvalue problem. Denoting by A the stiffness matrix and by M the mass matrix, the discrete modes satisfy

$$\mathbf{A}\underline{u} = k_h^2 \mathbf{M}\underline{u} \tag{1.16}$$

where k_h^2 are the discrete eigenvalues and k_h is the associated numerical wavenumber. The dispersion error is therefore interpreted as the difference between the spectra of the continuous and discrete operators, expressed in terms of wavenumbers.

1.4.3 Dispersion Curve and Numerical Results

The accuracy of the method is assessed by introducing the dimensionless resolution parameter h/λ, where h denotes the characteristic mesh size and λ the wavelength. The numerical dispersion is represented through the dispersion curve, defined as the ratio k_h/k plotted as a function of h/λ. The generalized eigenvalue problem (1.16) is solved in MATLAB for a sequence of meshes with $N \in \{10,\ 20,\ 40,\ 80,\ 160\}$ elements. For each discrete eigenvalue k_h^2, the corresponding exact wavenumber k is identified analytically and the ratio k_h/k is computed. Only modes with $h/\lambda < 0.5$ are retained to exclude poorly resolved optical branches; in practice, the data extend up to $h/\lambda \approx 0.36$ due to the limited number of available eigenvalues on the coarser meshes. The resulting dispersion curve is shown in figure 1.4.

1.4.4 Physical Interpretation

The dispersion curve in figure 1.4 exhibits a monotonically decreasing trend, in full agreement with the theoretical prediction for $\mathbb{P}_1$ Galerkin approximations and three distinct resolution regimes can be identified.

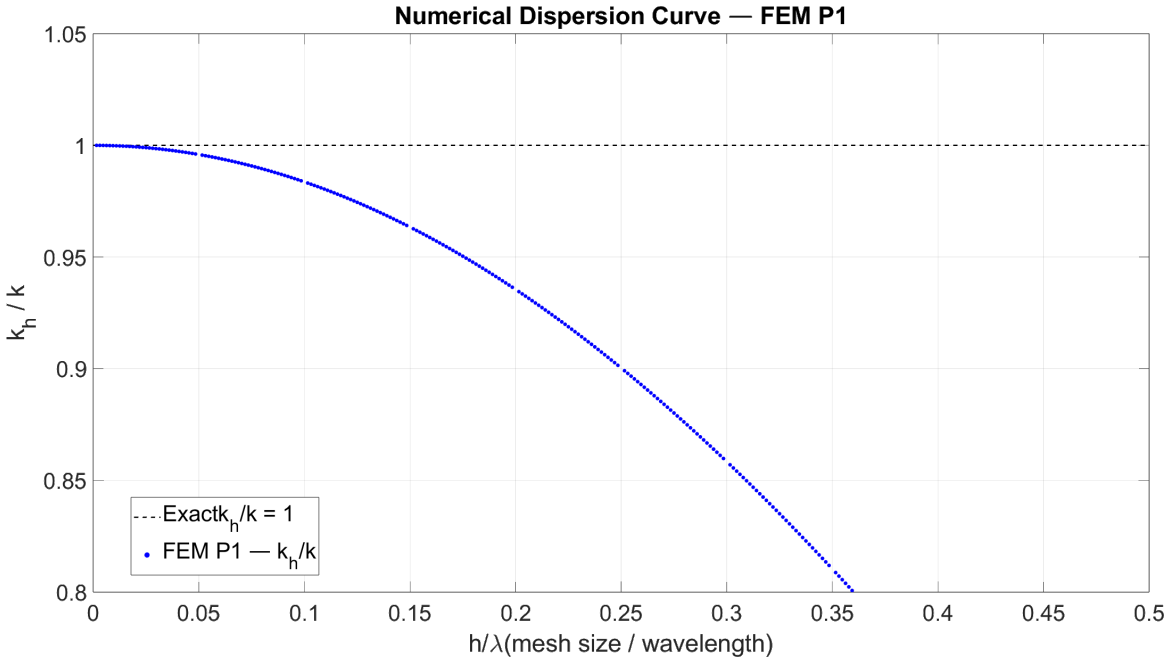


Figure 1.4: Numerical dispersion curve k_h/k as a function of the resolution parameter h/λ for the $\mathbb{P}_1$ FEM. The dashed line at $k_h/k = 1$ represents the exact (non-dispersive) behavior. As $h/\lambda \to 0$ the FEM solution converges to the exact wavenumber, while coarser meshes produce an increasingly large underestimation of k

Fine meshes — $h/\lambda \to 0$ As the mesh is refined, $k_h/k \to 1$ and the FEM reproduces the continuous wavenumber with negligible error. Below $h/\lambda \approx 0.05$, corresponding to roughly 20 or more elements per wavelength, the dispersion error is practically zero and the numerical phase velocity coincides with the physical one.

Intermediate resolution For $h/\lambda = 0.1$, i.e., approximately 10 elements per wavelength, the ratio is $k_h/k \approx 0.903$, corresponding to a dispersion error of about 1.7%. This level of resolution is commonly considered a practical minimum for engineering applications where moderate accuracy is acceptable.

Coarse meshes — large h/λ At $h/\lambda = 0.25$, corresponding to only 4 elements per wavelength, the ratio drops to $k_h/k \approx 0.90$, meaning a 10% underestimation of the physical wavenumber. In this regime, the FEM underestimates k_h and consequently overestimates the phase velocity, a well-known artifact of low-order Galerkin discretizations. This explains qualitatively the phase shift and amplitude distortion already observed on the coarse mesh in section 1.1.

General rule of thumb The analysis confirms that at least 10 elements per wavelength ($h/\lambda \leq 0.1$) are required to keep the dispersion error below 2% and that 20 or more elements per wavelength ($h/\lambda \leq 0.05$) are needed for high-fidelity simulations. These thresholds are consistent with the standard engineering guidelines for $\mathbb{P}_1$ FEM applied to wave propagation problems.

1.5 Perfectly Matched Layer (PML)

In order to reduce spurious reflections at the right boundary, an alternative to the absorbing (impedance) boundary condition at $x = 1$ is considered. The computational domain is extended by adding an absorbing layer, referred to as a Perfectly Matched Layer (PML), in the region $x \in (1, 1 + L_{\text{PML}})$. The PML is designed to damp outgoing waves without reflecting them back into the physical domain $\Omega = (0, 1)$.

1.5.1 Damping Profile

The attenuation mechanism is introduced through a spatially varying damping function $\sigma(x)$. In order to achieve effective absorption while minimizing numerical reflections due to abrupt coefficient variations, $\sigma(x)$ is required to increase smoothly from zero at the interface $x = 1$. A parabolic profile is adopted, defined as

$$\sigma(x) = \begin{cases} 0 & \text{for } x \leq 1 \\ \sigma_{\max} \left(\dfrac{x-1}{L_{\text{PML}}} \right)^2 & \text{for } x > 1 \end{cases} \tag{1.17}$$

1.5.2 Complex Stretching and Modified Coefficients

The PML formulation is obtained by introducing the complex stretching function

$$s(x) = 1 - i\frac{\sigma(x)}{\omega} \tag{1.18}$$

Within the layer, the Helmholtz model is recast by replacing the physical parameters ρ and μ with complex-valued equivalents,

$$\tilde{\rho}(x) = \rho(x)s(x), \qquad \tilde{\mu}(x) = \frac{\mu(x)}{s(x)} \tag{1.19}$$

This choice preserves the structure of the governing equation while introducing attenuation through complex coefficients.

1.5.3 Impedance Matching and Exponential Decay

A key property of the PML is that it is perfectly matched at the interface $x = 1$. In particular, the characteristic impedance in the PML region satisfies

$$Z_{\text{PML}} = \sqrt{\tilde{\rho}(x)\tilde{\mu}(x)} = \sqrt{\rho(x)s(x)\frac{\mu(x)}{s(x)}} = \sqrt{\rho(x)\mu(x)} = Z \tag{1.20}$$

which shows that the impedance is continuous across the interface. This condition implies, at the continuous level, perfect transmission and therefore the absence of reflected waves at $x = 1$. On the other hand, the local wavenumber becomes complex. Defining $k = \omega\sqrt{\rho/\mu}$ as the physical wavenumber, the PML wavenumber is obtained as

$$k_{\text{PML}}(x) = \omega\sqrt{\frac{\tilde{\rho}(x)}{\tilde{\mu}(x)}} = \omega\sqrt{\frac{\rho(x)s(x)}{\mu(x)/s(x)}} = \omega\sqrt{\frac{\rho(x)}{\mu(x)}}s(x) = ks(x) \tag{1.21}$$

Since $s(x) = 1 - i\sigma(x)/\omega$ has a negative imaginary part for $\sigma(x) \geq 0$, the transmitted wave in the layer experiences an exponential decay in amplitude as x increases, leading to strong attenuation before reaching the truncated boundary.

1.5.4 Boundary Conditions on the Extended Domain

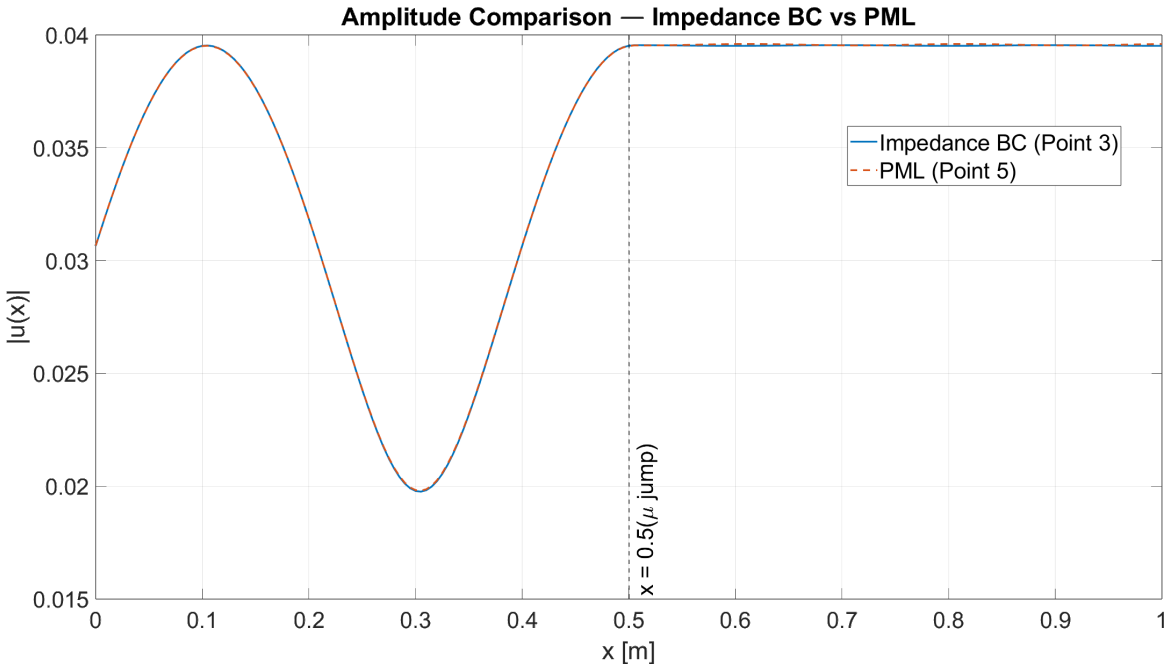
The excitation at the left boundary is kept unchanged and the Neumann condition is imposed at $x = 0$. The right boundary is moved to the end of the PML, located at $x = 1 + L_{\text{PML}}$, where a homogeneous Dirichlet condition is prescribed, $u(1+L_{\text{PML}}) = 0$. This truncation is justified by the assumption that the outgoing wave has been sufficiently damped by the PML and is therefore negligible at the outer boundary.

1.5.5 Numerical Results and Comparison

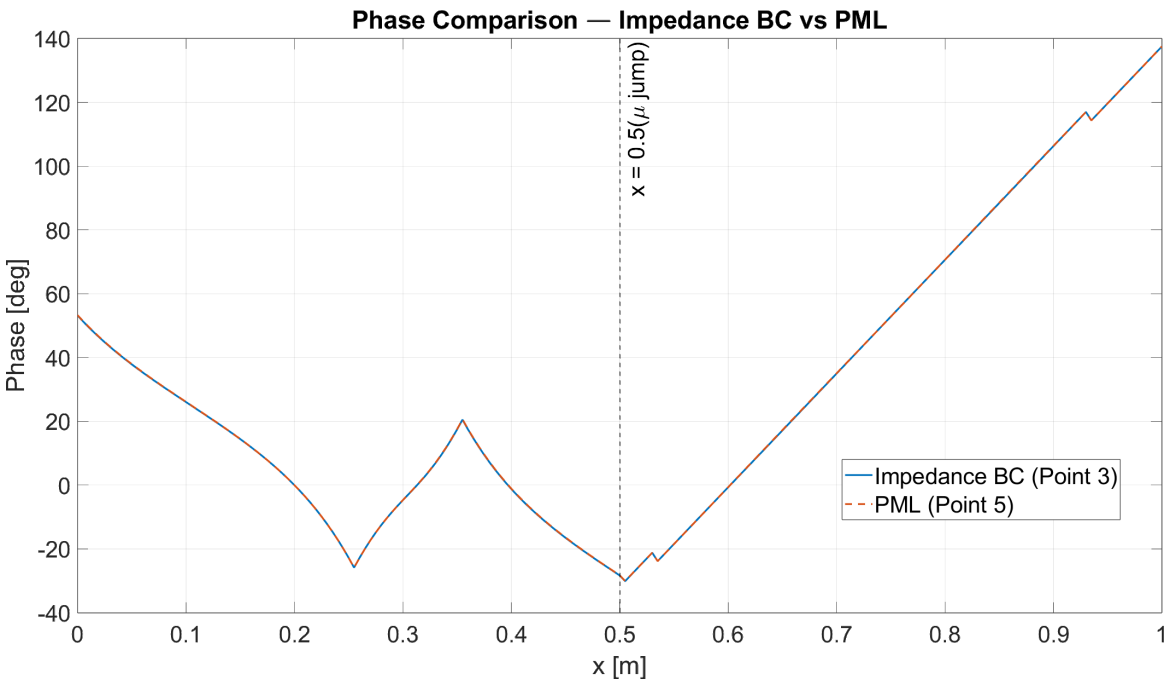
The PML approach is validated by comparing its solution against the impedance boundary condition result obtained in chapter 1.3, for the same physical setup: $\mu(x)$ as in equation (1.12), $\rho = 1$, $g_N = 1$, $\omega = 5\pi$ and $N = 200$ elements in the physical domain $(0, 1)$. The PML layer extends over $x \in (1, 1.2)$ with $L_{\text{PML}} = 0.2$ and $\sigma_{\text{max}} = 100$, discretized with an additional 40 elements, for a total of 240 degrees of freedom on the extended domain. Figure 1.5 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ restricted to the physical domain $(0, 1)$, comparing the two methods.

1.5.6 Physical Interpretation

The near-perfect agreement between the two solutions confirms that both methods correctly model the same physical phenomenon, a perfectly absorbing right boundary, through two fundamentally different strategies. The impedance boundary condition enforces the non-reflecting constraint analytically at a single point $x = 1$, while the PML distributes the attenuation numerically over an extended layer. The quasi-perfect overlap in both amplitude and phase throughout the physical domain validates the correctness of both implementations. The only visible discrepancy appears in the amplitude for $x > 0.5$, where the PML curve is marginally higher than the impedance BC result. This is a well-known discrete artifact: despite the impedance-matching property (1.20) holding exactly at the continuous level, the discretization introduces a small spurious reflection at the interface $x = 1$ whose magnitude depends on the choice of σ_{max} and L_{PML}. Increasing either parameter reduces the residual reflection, at the cost of a larger computational domain.



(a) Amplitude $|u(x)|$. Both curves overlap throughout $(0, 1)$: the oscillatory interference pattern for $x < 0.5$ and the smooth transmitted field for $x > 0.5$ are identical in the two approaches. A slight overestimation of the PML curve (dashed) is visible for $x > 0.5$, attributable to a small discrete numerical reflection at the PML interface



(b) Unwrapped phase $\arg(u(x))$. The irregular non-monotone behavior for $x < 0.5$ and the linear growth for $x > 0.5$ are virtually indistinguishable between the impedance BC and the PML solutions

Figure 1.5: Comparison of the FE solution obtained with the impedance boundary condition (Point 3, section 1.3) and the PML formulation (Point 5, section 1.5), restricted to the physical domain $\Omega = (0, 1)$

2

Homework 2

2.1 Equivalence to the Second-Order Wave Equation

The linearized shallow water equations govern the propagation of surface waves in a shallow domain $(x,t) \in (0,L) \times (0,T)$. The equivalence between the first-order coupled system and the classical second-order wave equation is established by eliminating one of the two unknowns.

2.1.1 Strong Formulation

The governing system reads:

$$\begin{cases} \eta_t(x,t) + q_x(x,t) = 0 \\ q_t(x,t) + gH\eta_x(x,t) = 0 \end{cases} \tag{2.1}$$

where $\eta(x,t)$ is the free surface elevation, $q(x,t)$ is the discharge, H is the mean water depth, and g is the gravitational acceleration. The system is supplemented by periodic boundary conditions:

$$\eta(0,t) = \eta(L,t), \quad q(0,t) = q(L,t) \tag{2.2}$$

2.1.2 Decoupling the System

The discharge variable $q(x,t)$ is eliminated by cross-differentiation. Differentiating the first equation of (2.1) with respect to time t:

$$\eta_{tt}(x,t) + q_{xt}(x,t) = 0 \quad \Longrightarrow \quad \eta_{tt}(x,t) = -q_{xt}(x,t) \tag{2.3}$$

Differentiating the second equation of (2.1) with respect to space x:

$$q_{tx}(x,t) + gH\eta_{xx}(x,t) = 0 \quad \Longrightarrow \quad q_{tx}(x,t) = -gH\eta_{xx}(x,t) \tag{2.4}$$

Assuming sufficient regularity, specifically $q \in C^2((0,L) \times (0,T))$, Schwarz's theorem yields $q_{xt} = q_{tx}$. Substituting (2.4) into (2.3) results in the standard second-order wave equation:

$$\eta_{tt}(x,t) = gH\eta_{xx}(x,t) \quad \text{for } (x,t) \in (0,L) \times (0,T) \tag{2.5}$$

2.1.3 Determination of Suitable Boundary Conditions

The periodic boundary conditions on the coupled variables (η, q) must be translated into conditions solely on η and its derivatives. The first condition is already explicit:

$$\eta(0,t) = \eta(L,t) \tag{2.6}$$

To convert the second condition $q(0,t) = q(L,t)$, differentiation with respect to t gives $q_t(0,t) = q_t(L,t)$. From the second equation of (2.1), $q_t = -gH\eta_x$. Evaluating at both boundaries and substituting:

$$-gH\eta_x(0,t) = -gH\eta_x(L,t) \quad \Longrightarrow \quad \eta_x(0,t) = \eta_x(L,t) \tag{2.7}$$

2.1.4 Equivalence

The first-order system (2.1) is therefore fully equivalent to the second-order acoustic wave equation:

$$\eta_{tt}(x,t) = gH\eta_{xx}(x,t), \quad (x,t) \in (0,L) \times (0,T) \tag{2.8}$$

subject to the periodic boundary conditions $\eta(0,t) = \eta(L,t)$ and $\eta_x(0,t) = \eta_x(L,t)$, derived in (2.6) and (2.7) respectively.

2.2 Conservation of Energy

The conservation of mechanical energy under periodic boundary conditions is established by showing that the time derivative of the prescribed energy functional vanishes identically.

2.2.1 Definition of the Energy Functional

The energy functional of the shallow water system over the spatial domain $(0, L)$ is defined as:

$$E(t) = \int_0^L \left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \tag{2.9}$$

where the first term represents the potential energy associated with the free surface elevation η, and the second term represents the kinetic energy associated with the discharge q. To prove conservation, it suffices to show that $\frac{dE}{dt} = 0$.

2.2.2 Time Derivative of the Energy

Assuming sufficient smoothness of $\eta(x,t)$ and $q(x,t)$, the Leibniz rule permits the exchange of differentiation and integration:

$$\frac{dE}{dt} = \frac{d}{dt}\int_0^L \left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx = \int_0^L \frac{\partial}{\partial t}\left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \tag{2.10}$$

Applying the chain rule to compute the partial time derivatives of the integrand yields:

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H} q q_t \right) dx \tag{2.11}$$

2.2.3 Substitution of the Strong Formulation

Recalling the linearized shallow water equations (2.1):

$$\begin{cases} \eta_t = -q_x \\ q_t = -gH\eta_x \end{cases} \tag{2.12}$$

Substituting these expressions for η_t and q_t into (2.11):

$$\frac{dE}{dt} = \int_0^L \left(g\eta(-q_x) + \frac{1}{H} q(-gH\eta_x) \right) dx \tag{2.13}$$

2.2.4 Algebraic Simplification and the Product Rule

Factoring out the common constant $-g$, the integral simplifies to:

$$\frac{dE}{dt} = -g \int_0^L (\eta q_x + q\eta_x)\, dx \tag{2.14}$$

The integrand is recognized as the exact spatial derivative of the product ηq via the product rule $\frac{\partial}{\partial x}(\eta q) = \eta_x q + \eta q_x$. Therefore:

$$\frac{dE}{dt} = -g \int_0^L \frac{\partial}{\partial x}(\eta q) dx \tag{2.15}$$

2.2.5 Application of the Fundamental Theorem of Calculus

Integrating the exact derivative over $[0, L]$ yields the evaluation of the boundary terms:

$$\frac{dE}{dt} = -g \Big[\eta(x,t)q(x,t) \Big]_{x=0}^{x=L} = -g\left(\eta(L,t)q(L,t) - \eta(0,t)q(0,t) \right) \tag{2.16}$$

2.2.6 Application of Boundary Conditions

The periodic boundary conditions prescribe $\eta(0,t) = \eta(L,t)$ and $q(0,t) = q(L,t)$. Substituting these into the boundary evaluation:

$$\eta(L,t)q(L,t) - \eta(0,t)q(0,t) = \eta(L,t)q(L,t) - \eta(L,t)q(L,t) = 0 \tag{2.17}$$

2.2.7 Energy Conservation

Therefore, the time derivative of the energy is exactly zero:

$$\frac{dE}{dt} = 0 \tag{2.18}$$

This rigorously demonstrates that the energy $E(t)$ defined in (2.9) is conserved for all $t \in (0, T)$ under the prescribed periodic boundary conditions.

2.3 Weak Formulation and Spectral Element Discretization

A variational formulation of the linearized shallow water system is derived and subsequently discretized using the Spectral Element Method (SEM), leading to a global first-order block ODE system suitable for time integration.

2.3.1 Function Spaces

The appropriate function space is defined to accommodate the periodic boundary conditions $\eta(0,t) = \eta(L,t)$ and $q(0,t) = q(L,t)$. Solutions are sought in the space of periodic functions with square-integrable first derivatives:

$$V = H^1_{\text{per}}(0,L) = \{v \in H^1(0,L) : v(0) = v(L)\}$$

For any $t \in (0,T)$, it is assumed that $\eta(\cdot,t) \in V$ and $q(\cdot,t) \in V$. Test functions are drawn from the same space: $v, w \in V$.

2.3.2 Weak Formulation

Starting from the strong form of the system:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = 0 \end{cases} \tag{2.19}$$

each equation is multiplied by a test function and integrated over the domain $(0,L)$:

$$\int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \quad \forall v \in V$$

$$\int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \quad \forall w \in V$$

Boundary Treatment Applying integration by parts to the convective term in the first equation:

$$\int_0^L q_x v dx = [qv]_0^L - \int_0^L q v_x dx$$

Since both q and v belong to V, the boundary term vanishes exactly by the periodicity constraint:

$$[qv]_0^L = q(L)v(L) - q(0)v(0) = 0$$

For the continuous Galerkin SEM with periodic basis functions, it is standard to retain the spatial derivatives on the trial functions. The weak formulation therefore reads: find $\eta, q \in V$ such that for all $v, w \in V$,

$$\begin{cases} \displaystyle\int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \\ \displaystyle\int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \end{cases} \tag{2.20}$$

2.3.3 Spectral Element Discretization

Mesh and Element Partition The domain $\Omega = (0, L)$ is partitioned into N_e non-overlapping elements $\Omega_e = [x_{e-1}, x_e]$. Each element is mapped to the reference element $[-1, 1]$ via an affine transformation.

Basis Functions A finite-dimensional subspace $V_h \subset V$ is introduced. In SEM, the basis functions $\phi_j(x)$ are piecewise continuous polynomials of degree N, specifically Lagrange polynomials constructed over Gauss-Lobatto-Legendre (GLL) nodes. The continuous solutions are approximated as:

$$\eta(x,t) \approx \eta_h(x,t) = \sum_{j=1}^{N_{\text{dof}}} \eta_j(t)\phi_j(x), \qquad q(x,t) \approx q_h(x,t) = \sum_{j=1}^{N_{\text{dof}}} q_j(t)\phi_j(x)$$

where N_{dof} is the total number of global degrees of freedom, with the periodic assembly mapping the last node to the first.

2.3.4 Galerkin System and Matrix Assembly

Substituting the discrete approximations and choosing $v = w = \phi_i(x)$, the semi-discrete system becomes:

$$\sum_j \dot{\eta}_j \int_0^L \phi_j \phi_i dx + \sum_j q_j \int_0^L \phi_j' \phi_i dx = 0$$

$$\sum_j \dot{q}_j \int_0^L \phi_j \phi_i dx + gH \sum_j \eta_j \int_0^L \phi_j' \phi_i dx = 0$$

The global mass matrix **M** and the first-derivative matrix **D** are defined by:

$$M_{ij} = \int_0^L \phi_i \phi_j dx, \qquad D_{ij} = \int_0^L \phi_i \phi_j' dx$$

Note on GLL Quadrature In SEM, integrals are evaluated using GLL quadrature. The discrete orthogonality property of the Lagrange basis at the GLL nodes renders **M** diagonal (lumped), which is advantageous for time integration. Introducing the global coefficient vectors:

$$\underline{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \vdots \\ \eta_{N_{\text{dof}}}(t) \end{bmatrix}, \qquad \underline{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_{N_{\text{dof}}}(t) \end{bmatrix}$$

the semi-discrete system takes the compact matrix form:

$$\mathbf{M}\dot{\underline{\eta}}(t) + \mathbf{D}\underline{q}(t) = \underline{0} \tag{2.21}$$

$$\mathbf{M}\dot{\underline{q}}(t) + gH\mathbf{D}\underline{\eta}(t) = \underline{0} \tag{2.22}$$

2.3.5 Final Block System

Stacking the unknown vectors into the global state vector $\underline{U} = \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix}$, the coupled system (2.21)-(2.22) is cast into the block matrix form:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix} \begin{bmatrix} \dot{\underline{\eta}} \\ \dot{\underline{q}} \end{bmatrix} = - \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix} \tag{2.23}$$

Defining the global block matrices:

$$\mathcal{M} = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix}, \qquad \mathcal{S} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \tag{2.24}$$

the target semi-discrete first-order ODE system is recovered:

$$\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \mathbf{0} \tag{2.25}$$

Here, $\mathcal{M}$ is symmetric and positive definite (diagonal under GLL quadrature), while $\mathcal{S}$ encodes the anti-symmetric spatial derivative operator scaled by the physical parameters.

2.4 MATLAB Solver

The global semi-discrete system (2.25) is integrated in time using a θ-method time integrator. The implementation is validated against the known exact solution through an L^2 error analysis and a spectral convergence study.

2.4.1 Problem Parameters and Exact Solution

The physical domain is set to $L = 1\,\mathrm{m}$, $H = 1\,\mathrm{m}$, and $g = 9.81\,\mathrm{m/s^2}$, yielding the wave speed $c = \sqrt{gH} = \sqrt{9.81}$ m/s. The simulation runs until the final time $T = 0.5\,\mathrm{s}$. The initial conditions are:

$$\eta(x,0) = e^{-50(x-0.5)^2}, \qquad q(x,0) = c\eta(x,0)$$

These initial data are compatible with a purely rightward-propagating wave. Consequently, the exact solution for the free surface elevation is a rigid translation of the initial profile:

$$\eta_{\text{exact}}(x,t) = \eta(x - ct, 0)$$

Under periodic boundary conditions, the argument $(x - ct)$ is evaluated modulo L.

2.4.2 Crank-Nicolson Time Integration Scheme

The global semi-discrete system derived in section 2.3 is a first-order ODE of the form:

$$\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \mathbf{0} \tag{2.26}$$

Time integration is performed using the θ-method. The choice $\theta = \frac{1}{2}$ corresponds to the Crank-Nicolson scheme, which is second-order accurate in time and unconditionally stable for the conservative system (2.26). This scheme is applied directly to the first-order block system, consistently with the formulation derived in section 2.3.

Discretization Approximating the time derivative by a first-order finite difference and evaluating the spatial operator at the weighted average $\theta\underline{U}^{k+1} + (1-\theta)\underline{U}^k$ with $\theta = \frac{1}{2}$ yields:

$$\mathcal{M}\frac{\underline{U}^{k+1} - \underline{U}^k}{\Delta t} + \mathcal{S}\left[\frac{1}{2}\underline{U}^{k+1} + \frac{1}{2}\underline{U}^k\right] = \mathbf{0} \tag{2.27}$$

Linear System per Time Step Collecting all terms involving the unknown $\underline{U}^{k+1}$ on the left-hand side, equation (2.27) is rearranged into the linear system:

$$\left(\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^{k} \tag{2.28}$$

At each time step, equation (2.28) constitutes a linear system for the unknown state vector $\underline{U}^{k+1}$. The system matrix $\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}$ depends only on the mesh and Δt, and may therefore be factorized once outside the time loop to optimize the solver. The right-hand side is updated at each step using the current state $\underline{U}^{k}$.

2.4.3 Computation of the L^2 Error Norm

The accuracy of the numerical scheme is quantified by the L^2 error norm $\|\eta_{\text{exact}} - \eta_h\|_{L^2(0,L)}$ evaluated at the final time T:

$$\|\eta_{\text{exact}}(\cdot,T) - \eta_h(\cdot,T)\|_{L^2(0,L)} = \left(\int_0^L \left|\eta_{\text{exact}}(x,T) - \eta_h(x,T)\right|^2 dx\right)^{1/2}$$

In the finite element framework, this integral is evaluated using the global mass matrix $\mathbf{M}$. Letting $\underline{e} = \underline{\eta}_{\text{exact}}(T) - \underline{\eta}_h(T)$ denote the vector of nodal errors at $t = T$, the discrete L^2 error is computed as:

$$E_{L^2} = \sqrt{\underline{e}^{\mathrm{T}}\mathbf{M}\underline{e}}$$

2.4.4 MATLAB Implementation Structure

The MATLAB code is modularized to manage the periodic assembly and the block time integration efficiently.

- Mesh Generation: Define the number of elements N_e and polynomial degree N. Generate the GLL nodes and weights. Construct the global mesh coordinates.
- Periodic Mapping: Construct a connectivity array that identifies the last global node with the first (`global_dof(end) = 1`), enforcing $\eta(0,t) = \eta(L,t)$ and $q(0,t) = q(L,t)$.
- Matrix Assembly: Loop over each element to construct local mass ($\mathbf{M}_e$) and first-derivative ($\mathbf{D}_e$) matrices using GLL quadrature. Assemble into global sparse matrices $\mathbf{M}$ and $\mathbf{D}$ via the periodic mapping. Owing to GLL quadrature, $\mathbf{M}$ is diagonal. Form the block matrices $\mathcal{M}$ and $\mathcal{S}$ as in (2.24).
- Initial Conditions: Evaluate $\eta_0(x)$ and $q_0(x)$ at the global nodes. Assemble the initial global state vector $\underline{U}^0 = [\underline{\eta}_0; \mathbf{q}_0]$.
- Time Integration (θ-Method): Pre-factorize the system matrix $\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}$ outside the loop. At each step k, evaluate the right-hand side $(\mathcal{M} - \frac{\Delta t}{2}\mathcal{S})\underline{U}^k$ and solve (2.28) for $\underline{U}^{k+1}$.
- Error Analysis: At $t = T$, extract $\underline{\eta}_h(T)$ from the state vector $\underline{U}$. Compute the exact nodal vector $\underline{\eta}_{\text{exact}}(T)$, accounting for periodicity via modular arithmetic. Evaluate the L^2 error using the assembled mass matrix.

A significant advantage of the variational formulation is that boundary conditions are incorporated naturally. The boundary integral in the variational form vanishes under periodicity without any explicit enforcement at $x = 0$ or $x = L$.

2.4.5 Numerical Results and Wave Propagation Profile

The numerical solver is executed with physical parameters $L = 1\,\text{m}$, $H = 1\,\text{m}$, $g = 9.81\,\text{m/s}^2$, final time $T = 0.5\,\text{s}$, and polynomial degree $N = 4$.

Wave Propagation — Periodic Rightward Pulse Figure **??** shows the free surface elevation at the final time step. The initial Gaussian pulse, originally centered at $x = 0.5$, propagates rightward at speed $c = \sqrt{gH} \approx 3.13\,\text{m/s}$. At $T = 0.5\,\text{s}$, the wave has traveled a total distance of approximately $1.56\,\text{m}$; since $L = 1\,\text{m}$, the pulse has wrapped around the domain, confirming correct enforcement of the periodic connectivity. The SEM approximation overlays the exact analytical solution with no visible phase lag (numerical dispersion) or amplitude decay (numerical dissipation), verifying the energy-preserving character of both the spatial discretization and the Crank-Nicolson time integrator.

2.4.6 Spectral Convergence Analysis

Theoretical convergence rates In the continuous Galerkin SEM framework, the domain is discretized with piecewise polynomial basis functions of degree N. For a sufficiently smooth initial condition, the spatial discretization error is expected to converge as:

$$E_{\text{spatial}} \approx C h^{N+1} \tag{2.29}$$

where $h = L/N_e$ is the element size and C is a constant independent of h. For polynomial degree $N = 4$, the theoretical spatial convergence rate is $\mathcal{O}(h^5)$, implying that halving h reduces the spatial error by a factor of $2^5 = 32$. The experimental order of convergence between two successive refinements is computed as:

$$r = \frac{\ln(E_{\text{coarse}}/E_{\text{fine}})}{\ln(h_{\text{coarse}}/h_{\text{fine}})} = \frac{\ln(E_{\text{coarse}}/E_{\text{fine}})}{\ln(N_{e,\text{fine}}/N_{e,\text{coarse}})} \tag{2.30}$$

Balancing spatial and temporal errors When evaluating convergence of the complete numerical scheme, both spatial and temporal error contributions must be considered:

$$E_{\text{total}} \leq E_{\text{spatial}} + E_{\text{temporal}} \approx C_1 h^{N+1} + C_2 \Delta t^2 \tag{2.31}$$

The Crank-Nicolson scheme is second-order accurate in time ($\mathcal{O}(\Delta t^2)$). Since the spatial scheme converges at the much faster rate $\mathcal{O}(h^5)$, the spatial error drops rapidly with mesh refinement. If Δt is not sufficiently small, E_{temporal} will dominate and the experimentally observed convergence rate will plateau well below the expected value of 5.0. A sufficiently small time step (e.g., $\Delta t \leq 10^{-4}$) must therefore be maintained to isolate the spectral spatial convergence on a log-log plot.

Numerical results The L^2 error norm is computed for sequentially refined meshes ($N_e = 10, 20, 40, 80$) at fixed polynomial degree $N = 4$ and a sufficiently small time step.

Spectral Convergence — $\mathcal{O}(h^5)$ Rate Figure **??** plots the experimental L^2 errors against N_e on a logarithmic scale. The numerical slope runs parallel to the $\mathcal{O}(h^5)$ reference line, yielding an experimental convergence rate of $r \approx 5.0$ between successive mesh refinements. This confirms the theoretical prediction (2.29) and validates the correct assembly of the SEM matrices and the periodic boundary enforcement.

2.5 Studies on Boundary Conditions and Friction

The effects of two physical modifications are investigated: replacing the periodic boundary conditions with reflecting walls, and introducing a linear friction term into the momentum equation. In each case, the consequences for energy conservation, the equivalent second-order equation, and the discrete formulation are derived.

2.5.1 Part A: Reflecting Wall Boundary Conditions

Strong formulation and equivalence The periodic boundary conditions are replaced by reflecting walls, defined by zero discharge at the domain endpoints:

$$q(0,t) = 0 \quad \text{and} \quad q(L,t) = 0 \tag{2.32}$$

To determine the equivalent condition on the second-order wave equation $\eta_{tt} = gH\eta_{xx}$, the second equation of the shallow water system, $q_t + gH\eta_x = 0$, is evaluated at the boundaries. Since $q(x,t) = 0$ at both walls for all t, the time derivative q_t also vanishes there. Therefore:

$$-gH\eta_x(0,t) = 0 \quad \Longrightarrow \quad \eta_x(0,t) = 0, \qquad -gH\eta_x(L,t) = 0 \quad \Longrightarrow \quad \eta_x(L,t) = 0 \tag{2.33}$$

A homogeneous Dirichlet condition on the discharge q is thus physically and mathematically equivalent to a homogeneous Neumann condition on the free surface elevation η. Waves reflect off the boundaries without phase inversion.

Effect on energy conservation Revisiting the time derivative of the energy functional $E(t)$ from section 2.2, integration leads to the boundary evaluation:

$$\frac{dE}{dt} = -g\Big[\eta(x,t)q(x,t)\Big]_{x=0}^{x=L} = -g\left(\eta(L,t)q(L,t) - \eta(0,t)q(0,t)\right) \tag{2.34}$$

Substituting the reflecting wall conditions $q(0,t) = q(L,t) = 0$, the boundary term vanishes:

$$\frac{dE}{dt} = 0 \tag{2.35}$$

Energy is therefore still perfectly conserved in a closed domain with reflecting walls; no energy enters or leaves the system through the boundaries.

Effect on the weak formulation and function spaces The function spaces must be modified. The periodicity constraint is dropped, and the free surface elevation belongs to $V = H^1(0,L)$. To enforce the Dirichlet condition on q strongly, the discharge must belong to $V_0 = H_0^1(0,L) = \{v \in H^1(0,L) : v(0) = v(L) = 0\}$. The test function w for the momentum equation is also drawn from V_0, so that the boundary terms in the variational formulation vanish by construction.

2.5.2 Part B: Friction

Modified strong formulation A linear friction term proportional to the discharge is introduced into the momentum equation with friction coefficient $\gamma > 0$:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = -\gamma q \end{cases} \tag{2.36}$$

Equivalence to the damped wave equation Differentiating the continuity equation with respect to time gives $\eta_{tt} = -q_{xt}$. Differentiating the momentum equation with respect to space x:

$$q_{tx} + gH\eta_{xx} = -\gamma q_x \tag{2.37}$$

Applying Schwarz's theorem ($q_{xt} = q_{tx}$) and substituting $\eta_{tt} = -q_{tx}$ and $-\eta_t = q_x$:

$$-\eta_{tt} + gH\eta_{xx} = \gamma\eta_t \quad \Longrightarrow \quad \eta_{tt} + \gamma\eta_t = gH\eta_{xx} \tag{2.38}$$

This is the Telegrapher's equation (damped wave equation). The term $\gamma\eta_t$ represents the physical attenuation of wave amplitude over time.

Effect on energy dissipation The time derivative of the total energy $E(t)$ defined in (2.9) is:

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H} q q_t \right) dx \tag{2.39}$$

Substituting the modified strong formulation (2.36):

$$\frac{dE}{dt} = \int_0^L \left(g\eta(-q_x) + \frac{1}{H} q(-gH\eta_x - \gamma q) \right) dx \tag{2.40}$$

Expanding and collecting terms yields:

$$\frac{dE}{dt} = -g \int_0^L (\eta q_x + q\eta_x) dx - \int_0^L \frac{\gamma}{H} q^2 dx \tag{2.41}$$

The first integral reduces to a boundary term via the product rule. Under either periodic or reflecting boundary conditions this boundary term vanishes, leaving:

$$\frac{dE}{dt} = -\frac{\gamma}{H} \int_0^L q^2 dx \leq 0 \tag{2.42}$$

Since $\gamma > 0$, $H > 0$, and $q^2 \geq 0$, the time derivative of the energy is strictly non-positive. Friction therefore monotonically dissipates the mechanical energy of the system.

Effect on the Galerkin discretization In the discrete setting, the friction term modifies exclusively the spatial block matrix $\mathcal{S}$ of the global first-order system (2.25). The friction contributes a scaled mass matrix to the $(2, 2)$ block of $\mathcal{S}$, corresponding to the discrete counterpart of the term $-\gamma q$ in the momentum equation. The modified block matrix is:

$$\mathcal{S}_{\text{friction}} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \gamma\mathbf{M} \end{bmatrix} \tag{2.43}$$

The block mass matrix $\mathcal{M}$ remains unchanged. The θ-method time integration scheme described in section 2.4 applies without modification: it suffices to replace $\mathcal{S}$ with $\mathcal{S}_{\text{friction}}$ in the linear system (2.28), yielding:

$$\left(\mathcal{M} + \frac{\Delta t}{2} \mathcal{S}_{\text{friction}} \right) \underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2} \mathcal{S}_{\text{friction}} \right) \underline{U}^k \tag{2.44}$$

This substitution preserves the complete structure of the time integrator, requiring only the reassembly of the system matrix prior to the time loop.

2.5.3 Numerical Results and Physical Analysis

Two isolated simulations are conducted to validate the theoretical derivations of this section. The first isolates the effect of reflecting wall boundary conditions (with $\gamma = 0$), while the second isolates the effect of linear friction under periodic boundary conditions. Both experiments use the finest mesh ($N_e = 80$, $N = 4$) and a sufficiently small time step to avoid temporal error contamination.

Analysis of reflecting walls The upper panel of figure **??** shows the solution profile under homogeneous Dirichlet boundary conditions on the discharge ($q = 0$ at $x = 0, L$). The initial Gaussian pulse, propagating rightward, encounters the boundary at $x = L$. Since the discharge is forced to zero at the wall, the equivalent second-order formulation enforces a homogeneous Neumann condition on η_x, producing a perfect in-phase reflection without phase inversion. The peak amplitude of the reflected wave exactly reaches unity, corroborating the theoretical proof from section 2.5: the system remains perfectly conservative, and no mechanical energy is lost at the wall.

Analysis of friction The lower panel of figure **??** depicts wave propagation under periodic boundaries with the linear friction term $q_t + gH\eta_x = -\gamma q$, using $\gamma = 2.0$. The damped wave (solid line) remains perfectly in phase with the undamped exact solution, confirming that the wave speed $c = \sqrt{gH}$ is unaffected by linear damping. However, the amplitude has decayed significantly below unity, visualizing the strictly non-positive energy derivative (2.42). This dissipative behavior is captured numerically by the modified block matrix $\mathcal{S}_{\text{friction}}$ defined in (2.43), which enters the Crank-Nicolson system (2.44) without any structural change to the time integrator.